Lab 53 Worksheet — Safety Procedures, ESD Control and Environmental Compliance

Name: ____ **Date:** __

Exam objective: Apply ESD, electrical and personal safety procedures, and comply with environmental requirements for disposal and equipment protection (Core 2 objectives 4.4 and 4.5).

Record as you go

Step	What you did	What you observed
1	Record what electrostatic discharge is and why it matters: a person can carry thousands of volts of static, and components are damaged at levels far below what a person can feel.	
2	Record the ESD controls: an anti-static wrist strap connected to earth, an anti-static mat, anti-static bags for storage, and controlling humidity since dry air increases static build-up.	
3	Record the handling rules: hold expansion cards by their edges, never touch gold contacts or chip pins, and equalise potential by touching the chassis before touching a component.	
4	Record the electrical safety rules: disconnect mains power before working inside a machine, hold the power button for 15 seconds to discharge, and never open a power supply or a CRT monitor.	
5	Record why a power supply and a CRT are singled out: both retain a lethal charge in their capacitors long after being unplugged.	
6	Record the fire safety rules: use a Class C extinguisher on an electrical fire, never water, and know where the extinguisher and the power cut-off are before you need them.	

Step	What you did	What you observed
7	Record the personal safety rules: correct lifting technique using the legs with a straight back, safety goggles when using compressed air, and an air filtration mask when cleaning dust or toner.	
8	Record the MSDS or SDS requirement: every hazardous material has a safety data sheet stating its hazards, its handling requirements and its disposal method, and it must be accessible to staff.	
9	Record the disposal requirements: batteries, toner cartridges, CRT monitors and electronic waste must all go to approved recycling and never into general waste.	
10	Record the environmental controls: temperature and humidity management, dust control with compressed air or an anti-static vacuum, and adequate ventilation.	
11	Record the power protection controls: a surge suppressor against spikes, and an uninterruptible power supply against sags, brownouts and outages, giving time for an orderly shutdown.	
12	Perform a hazard assessment of your actual workspace, listing every hazard found, its risk level and the specific control you would implement.	

Verification

Success criterion: Your procedure covers ESD, electrical, personal and environmental safety; it explicitly forbids opening a PSU and a CRT and states why; Class C is named for electrical fires; and the hazard assessment lists real hazards with a control for each.

- I completed every step in the lab.
- My result meets the success criterion above.

- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
